

Can also invert rational $X(z)$ with long division ³

Ex. $X(z) = \frac{1-z^{-1}}{1-\frac{1}{2}z^{-1}}$ $|z| > \frac{1}{2}$

$$\begin{array}{r} 1 - \frac{1}{2}z^{-1} \overline{) 1 - z^{-1}} \\ \underline{1 - \frac{1}{2}z^{-1}} \\ 0 - \frac{1}{2}z^{-1} \\ \underline{-\frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}} \\ 0 - \frac{1}{4}z^{-2} \\ \underline{-\frac{1}{4}z^{-2} + \frac{1}{8}z^{-3}} \\ 0 - \frac{1}{8}z^{-3} \\ \underline{-\frac{1}{8}z^{-3} + \frac{1}{16}z^{-4}} \\ 0 - \frac{1}{16}z^{-4} \\ \vdots \end{array}$$

$x[n] = \begin{matrix} 0 & n < 0 \\ 1 & n = 0 \\ -\frac{1}{2} & n = 1 \\ -\frac{1}{4} & n = 2 \\ -\frac{1}{8} & n = 3 \\ -\frac{1}{16} & n = 4 \\ \vdots & \end{matrix}$

$x[n] = 2\delta[n] - (\frac{1}{2})^n u[n]$

don't give any picture just explain why $2 * \delta[n]$ of n



The appearance of the term $2\delta[n]$ follows directly from **algebraic decomposition in the z -domain** and the **basic Z-transform pair for a constant**. No pictures are needed.

1. Start from the given transform

$$X(z) = \frac{1 - z^{-1}}{1 - \frac{1}{2}z^{-1}}, \quad |z| > \frac{1}{2}$$

2. Rewrite $X(z)$ using algebra (key step)

Express the numerator in terms of the denominator:

$$1 - z^{-1} = 2\left(1 - \frac{1}{2}z^{-1}\right) - 1$$

Substitute into $X(z)$:

$$X(z) = \frac{2\left(1 - \frac{1}{2}z^{-1}\right) - 1}{1 - \frac{1}{2}z^{-1}} = 2 - \frac{1}{1 - \frac{1}{2}z^{-1}}$$

This is the crucial point.

3. Apply standard inverse Z-transform pairs

1. Constant term

$$2 \xleftrightarrow{\mathcal{Z}^{-1}} 2\delta[n]$$

This is a fundamental result:

- A constant in $X(z)$ corresponds to a scaled impulse at $n = 0$.

2. Geometric term

$$\frac{1}{1 - \frac{1}{2}z^{-1}} \xleftrightarrow{\mathcal{Z}^{-1}} \left(\frac{1}{2}\right)^n u[n]$$

4. Final time-domain signal

$$x[n] = 2\delta[n] - \left(\frac{1}{2}\right)^n u[n]$$

5. Why exactly $2\delta[n]$? (Core intuition)

- The value of an impulse equals the **constant (direct-feedthrough) term** in $X(z)$.
- When long division produces a leading constant 2, that means:
 - There is an instantaneous response at $n = 0$
 - Its magnitude is exactly 2
- Therefore:

$$\text{constant } 2 \Rightarrow 2\delta[n]$$

One-line takeaway

Any constant term in $X(z)$ becomes a scaled impulse in time.

Here, the constant is 2, so the result must include $2\delta[n]$.



